Sahaj Bhadja

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Summary

Full-Stack Software Engineer with 2+ years building **React applications**, **REST APIs**, relational data systems, and automated cloud deployments for reliable, maintainable, user-centered production workflows.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, California*

- Architected a multi-tenant application platform using Azure SQL and Blob Storage for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through query and indexing optimization.
- Built FastAPI integrations across PostgreSQL, SQL Server, and cloud storage, processing 5K+ monthly jobs with validation, reconciliation, error handling, and CI/CD-backed releases at 99%+ measured production reliability.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, California*

- Built **React interfaces** and Python services for uploads, grading history, configuration, and live feedback, reducing user turnaround 60% while supporting reliable production workflows used by 150+ instructors.
- Designed concurrent processing with multithreading, secure file ingestion, AWS RDS, and **REST APIs**, evaluating 10K+ submissions daily through recoverable execution paths while maintaining 90%+ grading consistency reliably.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer, SecureSync *India*

- Engineered cross-platform microservices with automated VM builds and 20+ unit and integration suites using Jenkins, PyTest, and Mockito across Windows, Linux, and macOS production release environments reliably.
- Optimized object-oriented API integrations and AWS data flows, improving response times 30% and pipeline efficiency 68% while reducing technical debt 20% through reusable services and structured reviews.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern, VibrantChem *India*

- Implemented eight microservices spanning authentication, permissions, billing, resources, customers, leads, and notifications, reliably synchronizing 2M+ records through REST APIs and event-driven WebSocket updates across distributed production workflows.
- Built responsive React workflows and automated Docker deployments through Jenkins CI/CD, achieving 95%+ successful validation and cutting release time 50% through repeatable configurations, staged testing, and monitoring.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Built backend APIs for a cross-platform service-booking application covering bookings, time-slot selection, reservation tracking, and payment status updates across 50K+ records with consistently reliable sub-200-millisecond response times.
- Designed end-to-end system workflows, mapped 12+ backend integration points, and presented technical solutions to client stakeholders, reducing design errors 25% while supporting effective collaborative cross-functional implementation alignment.

Skills

Languages & Web: TypeScript, JavaScript, Python, Java, C++, SQL, HTML, CSS, React, Next.js, Angular, TailwindCSS
Backend & APIs: FastAPI, Flask, Node.js, Express.js, REST APIs, WebSockets, Microservices, OOP, JWT, Webhooks
Data & Cloud: SQL Server, Azure SQL, PostgreSQL, MySQL, MongoDB, Firebase, AWS, Azure, Docker, Kubernetes
Engineering: Git, Jenkins, CI/CD, Agile/Scrum, Unit Testing, Integration Testing, Code Reviews, Debugging, Monitoring, System Validation

Projects

HMIS - Hospital Management Information System Live | *TypeScript, MERN, ABDM APIs, Sentry*

- Architected a microservice-based healthcare platform integrating ABDM APIs, supporting 100+ daily appointment workflows and reducing patient-record retrieval by 2x through structured schemas, consent controls, and real-time monitoring.

FinNexus Live | *Next.js, TypeScript, Kubernetes, MongoDB*

- Built a TypeScript application supporting 50K+ users and 1M+ transactions with responsive interfaces, secure integrations, real-time synchronization, advanced filtering, and reliable containerized deployment workflows at production scale.

Big-Brain Live | *PostgreSQL, GCP, REST APIs, Clerk*

- Built secure document upload, management, and collaborative workflows for 200+ users with Clerk authentication, PostgreSQL-backed metadata, real-time summarization, and optimized retrieval pipelines that reduced review time 60%.